Example 8.29 If A and B are two events such that $P(A)=\frac{1}{4}$, $P(B)=\frac{1}{2}$ and $P(A \text{ and } B)=\frac{1}{8}$, find (i) $P(A \text{ or } B)$ (ii) $P(\text{not } A \text{ and not } B)$.

Solution (i) $P(A \text{ or } B) = P(A \cup B)$

$$= P(A)+P(B)-P(A \cap B)$$

$$P(A \text{ or } B) = \frac{1}{4}+\frac{1}{2}-\frac{1}{8} = \frac{5}{8}$$

(ii) $P(\text{not } A \text{ and not } B) = P(\bar{A} \cap \bar{B})$

$$= P\left(\overline{A \cup B}\right)$$

$$= 1-P\left(A \cup B\right)$$

$$P(\text{not } A \text{ and not } B) = 1-\frac{5}{8} = \frac{3}{8}$$

Example 8.30 A card is drawn from a pack of 52 cards. Find the probability of getting a king or a heart or a red card.

Solution Total number of cards = 52; $n(S)=52$

Let A be the event of getting a king card. $n\left(A\right)=4$

$$P\left(A\right) = \frac{n(A)}{n(S)} = \frac{4}{52}$$

Let B be the event of getting a heart card. $n\left(B\right)=13$

$$P\left(B\right) = \frac{n(B)}{n(S)} = \frac{13}{52}$$

Let C be the event of getting a red card. $n\left(C\right)=26$

$$P(C) = \frac{n(C)}{n(S)} = \frac{26}{52}$$

$$P(A \cap B) = P(\text{getting heart king}) = \frac{1}{52}$$

$$P(B \cap C) = P(\text{getting red and heart})) = \frac{13}{52}$$

$$P(A \cap C) = P(\text{getting red king}) = \frac{2}{52}$$

$$P(A \cap B \cap C) = P(\text{getting heart, king which is red}) = \frac{1}{52}$$

Suits of playing cards	Spade ♠	Heart ♥	Clavor ♣	Diamond ♦
Cards of each suit	A	A	A	A
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9
	10	10	10	10
	J	J	J	J
	Q	Q	Q	Q
	K	K	K	K
Set of playing cards in each suit	13	13	13	13

Therefore, required probability is

$$P(A \cup B \cup C) = P(A)+P(B)+P(C)-P(A \cap B)-P(B \cap C)-P(C \cap A)+P(A \cap B \cap C)$$

$$= \frac{4}{52}+\frac{13}{52}+\frac{26}{52}-\frac{1}{52}-\frac{13}{52}-\frac{2}{52}+\frac{1}{52} = \frac{28}{52} = \frac{7}{13}$$

Example 8.31 In a class of 50 students, 28 opted for NCC, 30 opted for NSS and 18 opted both NCC and NSS. One of the students is selected at random. Find the probability that

(i) The student opted for NCC but not NSS.

8 Stats and Probability.indd 328 15-12-2020 03:24:44 PM